When Background Noise Has Meaning

Environmental Sound Classification under Semantic Interference

Qihan Zhu SID 550489593 ELEC5305

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Abstract

Environmental recordings often contain more than one recognisable event. A classifier can fail because its acoustic representation is corrupted, or because a competing event captures its decision. This study compares a handcrafted 92-feature SVM, a compact log-Mel CNN and frozen zero-shot LAION-CLAP on all five official ESC-50 folds. Each target is tested with white noise, DEMAND background audio and same- or different-category ESC-50 distractors at +10, +5 and 0 dB. Identical mixtures and gain-matched clean controls are shared by all systems. Two activity-based power definitions expose a temporal-overlap confound. At +5 dB, the SVM is especially vulnerable to white noise, whereas CLAP retains strong performance under ordinary backgrounds but loses substantially more accuracy under overlapping meaningful events. On high-overlap targets, CLAP's semantic-minus-DEMAND accuracy difference is approximately -26 to -29 percentage points; the corresponding CNN difference is about -3 points and requires more cautious statistical interpretation. Distractor capture, full-class rankings and within-model representation shifts reveal structured failure patterns that aggregate accuracy alone misses. The contribution is a reproducible interference experiment and failure analysis, rather than a claim about the best architecture or the causal effect of semantics alone.

1 Research question and context

A single-label environmental-sound benchmark usually identifies a dominant event. Deployment audio can contain a dog bark and an alarm simultaneously. Equal interference power does not imply equal spectral structure, temporal overlap, recognisability or perceptual prominence. The main question is therefore: *How do different audio representations fail under ordinary acoustic noise and a competing meaningful event at controlled relative levels?* A second question asks whether an erroneous prediction identifies the distractor itself.

ESC-50 provides a small balanced benchmark with official folds [1, 2]. Spectrogram CNNs established a learned alternative to handcrafted descriptors [3]; augmentation has also been studied for environmental-sound classification [4]. A third paradigm, contrastive audio-language pretraining, permits classification through similarity to textual class descriptions [6]. These systems have very different external training histories. Their comparison here concerns deployment behaviour, not equal-data training efficiency.

Recent work motivates a closer examination of structured interference. PAT uses training-free cross-modal alignment and evaluates noisy inputs [7]. Mix-CLAP studies noise-aware audio-language models and efficiency [8]. RSA-Bench constructs complex acoustic backgrounds for speech perception and reasoning tasks [9]. That benchmark is related motivation, not a direct environmental-event classification baseline. This project does not reproduce those model-training programmes.

2 Dataset, models and validation

2.1 Official folds and data handling

ESC-50 contains 2,000 five-second recordings, 50 balanced classes and five predefined folds. All 50 classes are retained. For outer fold k, fold $(k \bmod 5) + 1$ supplies validation data and the remaining three folds supply selection training data. After selecting hyperparameters or training duration, a fresh model is fitted on the four non-test folds. Test labels are used only for evaluation. Feature standardisation and Mel normalisation are estimated from the applicable training partition.

The data audit identifies four source IDs spanning official folds, affecting nine target clips. Official assignments are preserved and documented; those sources are excluded from semantic-distractor eligibility. Consequently, the experiment should not be described as perfectly source-independent. The dataset authors also document partly class-dependent preprocessing [2]; clean benchmark performance is not evidence of unrestricted real-world generalisation.

2.2 DSP and classical baseline

The custom STFT uses a 1,024-sample periodic Hann window, a 512-sample hop and a 1,024-point FFT at 44.1 kHz. Frames start at sample zero; only the final incomplete frame is zero-padded. The output contains 513 complex one-

sided bins. No per-clip amplitude normalisation or pre-emphasis is applied. Agreement is checked against SciPy and against an actual MATLAB execution of the unmodified course `MyStft.m`. The latter comparison explicitly reconciles the supplied window, leading padding, frame offsets and full/one-sided spectra. On a real ESC-50 recording the relative complex-valued error is approximately 2.6×10^{-16}. This verifies corresponding frames, not identical boundary policies or default window helpers.

The handcrafted representation aggregates means and standard deviations of RMS, zero-crossing rate, spectral centroid, spread, roll-off, flux, 20 MFCCs and their deltas, giving 92 dimensions. An RBF SVM uses training-only standardisation. The original nine-candidate grid combines $C \in \{1, 10, 100\}$ with $\gamma \in \{\texttt{scale}, 0.001, 0.1\}$, selected by clean validation macro-F1. The implementation retains one-vs-rest decision scores and explicit tie handling.

2.3 Compact CNN and frozen CLAP

The CNN uses 64 Mel bands, log compression, and three convolution/normalisation/ ReLU/pooling blocks with 16, 32 and 64 channels. Temporal pooling and four frequency bins yield a 256-dimensional representation before the classifier. It has 36,258 trainable parameters. Adam uses learning rate 10^{-3}, weight decay 10^{-4}, batch size 32 and dropout 0.3. Training is capped at 40 epochs with patience 8; clean validation selects duration before a fresh four-fold refit. The core experiment uses seed 5305. Additional seeds are analysed separately.

CLAP uses the fixed `630k-audioset-best.pt` checkpoint, HTSAT-tiny audio encoder, RoBERTa text encoder and no fusion. Audio is resampled to 48 kHz and passes through the official int16 truncation-and-return convention exactly once. The prompt is `the sound of {label}`, replacing underscores with spaces. Class scores are cosine similarities between unit audio embeddings and the 50 unit prompt embeddings. Parameters are frozen; there is no ESC-50 classifier training, prompt selection or fine-tuning. Checkpoint SHA-256 starts `8053c977`; the complete hash is stored with the results. External pretraining overlap cannot be ruled out merely because inference is zero-shot.

3 Controlled interference and the power-definition change

Each target receives white Gaussian noise, a five-second segment from one of three DEMAND recordings (traffic, station and metro) [5], one same-coarse-category ESC-50 distractor and one different-coarse-category distractor. Semantic distractors have a different fine class and source recording and come from the target's held-out fold. Pairing is deterministic. Within each target/condition, the exact noise waveform, DEMAND segment or semantic pair is fixed across the three levels. Semantic events use zero relative offset.

Let $x[n]$ be the target and $d[n]$ the interferer. Activity masks contain samples covered by unwindowed frames whose mean-square energy is within 40 dB of the corresponding peak frame. Define $P(z; M) = |M|^{-1} \sum_{n \in M} z[n]^2$. For requested level L,

$$P_x = P(x; M_x), \tag{1}$$

$$P_d^{\text{own}} = P(d; M_d), \qquad P_d^{\text{target}} = P(d; M_x), \tag{2}$$

$$\alpha = \sqrt{P_x / (10^{L/10} P_d)}, \qquad y[n] = g\big(x[n] + \alpha d[n]\big), \tag{3}$$

$$g = \min\left(1, \frac{0.99}{\max_n |x[n] + \alpha d[n]|}\right). \tag{4}$$

The matching clean control is $gx[n]$. Common attenuation preserves the component-power ratio and avoids hard clipping. Achieved ratios are recomputed numerically rather than assumed from the gain formula.

Deviation from the proposal. The proposal specified the target-mask definition. In preliminary implementation, sparse distractors with little energy inside M_x required very large gains, even when their event occurred outside the target-active interval. The default was therefore changed to own-activity power. The originally proposed target-mask protocol was also run in full as a sensitivity analysis. The two ratios answer different measurement questions; neither is an automatic correction for temporal overlap, and the choice is not attributed to an unrecorded instructor instruction.

There are 24,000 own-protocol mixtures. Three target/distractor pairs have zero interferer energy on the target mask; their three levels are undefined under the target protocol and are excluded with recorded reasons, leaving 23,991 mixtures. Cross-protocol comparisons use common retained targets. The manifest records target/distractor identity, labels, categories, folds, hashes, seeds, offsets, gains, both measured power ratios, peak control and activity overlap $|M_x \cap M_d|/|M_x|$. The 0-dB condition is a deliberately ambiguous single-label stress test, not an ordinary equal-difficulty accuracy benchmark.

4 Outcomes and statistical treatment

Accuracy and macro-F1 are computed within each official test fold, with means and sample standard deviations across five folds. Macro-F1 retains all 50 class labels. For a semantic condition, distractor capture rate is the number of wrong predictions equal to the distractor divided by the number of errors. The unconditional capture fraction uses all trials instead. A reference of 1/49 assumes uniform choice among wrong labels; it is not an empirical null for class-dependent confusion.

The target–distractor margin subtracts the distractor score from the target score. Scores are SVM decision values, CNN logits or CLAP cosines and are interpreted only within a system. A negative margin does not guarantee that the distractor beats every other class. The CLAP diagnostic therefore also computes ranks and competing scores across all 50 prompts.

Paired bootstrap intervals resample target clips within official folds, retaining matched condition predictions, using 2,000 replicates and seed 530509. For these matched comparisons, accuracy differences pool the retained target trials; macro-F1 differences average the fold-specific values. They condition on the fitted models and do not incorporate refitting or all source-recording dependence. Exact McNemar comparisons use paired correctness. Holm correction covers the 12 +5-dB high-overlap comparisons across models, power definitions and semantic pairing types. These are post-baseline extensions, not retrospectively preregistered tests. Three CNN seeds separately assess training variability; five folds times three seeds are not 15 independent datasets.

5 Core results

Table 1: Clean five-fold results (percent; mean ± sample SD). Core CNN seed: 5305.

System	Accuracy	Macro-F1
SVM	48.90 ± 4.21	47.96 ± 4.07
CNN	67.90 ± 3.65	66.61 ± 3.91
CLAP	81.75 ± 2.58	79.49 ± 3.28

Table 2: Macro-F1 at +5 dB (percent; five-fold mean ± sample SD). Each cell uses its retained targets.

Power	Condition	SVM	CNN	CLAP
Own	White noise	2.03 ± 0.86	7.76 ± 1.54	62.91 ± 3.50
Own	DEMAND	16.13 ± 1.95	29.46 ± 4.19	71.49 ± 1.00
Own	Same category	22.47 ± 2.68	32.60 ± 2.26	48.86 ± 1.53
Own	Different category	21.34 ± 2.58	32.00 ± 3.38	45.16 ± 1.11
Target	White noise	2.03 ± 0.86	7.76 ± 1.54	62.92 ± 3.50
Target	DEMAND	16.15 ± 2.29	29.33 ± 4.14	71.64 ± 0.91
Target	Same category	21.14 ± 2.63	31.70 ± 1.88	47.76 ± 1.07
Target	Different category	20.87 ± 2.37	30.93 ± 4.19	42.81 ± 1.27

Table 1 establishes the three clean reference points. Table 2 shows +5-dB macro-F1 for each power definition; the complete level curves are in Figure 1. The SVM is severely affected by white noise, whereas meaningful distractors are less damaging in its aggregate results. This contradicts a universal assertion that semantic interference must always be harder than generic noise. CLAP is comparatively robust to ordinary backgrounds but experiences a much larger semantic-interference penalty.

Under the own protocol at +5 dB, CLAP's capture rates are approximately 67% for same-category and 60% for different-category distractors. The corresponding CNN values are about 31% and 28%, and SVM values about 19% and 16%. These are fractions of errors, not percentages of all targets. Capture and margin curves describe increasingly strong distractor competition as SIR falls (Figures 2 and 6). Both absolute accuracy and capture denominators must accompany this interpretation.

5.1 Overlap and matched comparisons

The whole-dataset CNN ranking between semantic interference and DEMAND is affected by the composition of low-overlap mixtures. Compare each semantic condition against DEMAND on precisely the same targets, stratified by

overlap. At overlap at least 0.8 and +5 dB, CNN semantic accuracy is about three percentage points lower. Pointwise paired bootstrap intervals exclude zero, but the Holm-adjusted exact McNemar p-values are approximately 0.12–0.15. The appropriate claim is a consistent observed direction requiring replication, rather than an established family-wise significant CNN effect.

CLAP's corresponding losses are much larger: own-protocol differences are -25.55 points (same category) and -28.53 points (different category), with paired 95% intervals [-28.28,-22.95] and [-31.20,-25.73]. The target protocol gives similarly large differences. Figure 3 separates these effects from the small, uncertain SVM differences. Stratified macro-F1 is also reported, but it can be depressed by missing classes; matched accuracy and class coverage are used to interpret stratum difficulty.

Changing power convention moderately lowers semantic performance but does not remove the main model-specific pattern. It should not be claimed that every fine ranking is invariant: small same-/different-category order changes can occur, including for the CNN at 0 dB. Both protocols and the retained-target intersection remain part of the result rather than selecting a preferred table.

6 Representation and competition diagnostics

Handcrafted displacement uses the relevant SVM training-fold scale and is normalised by the square root of group dimensionality. The eight groups are RMS, ZCR, centroid, spread, roll-off, flux, MFCCs and delta-MFCCs. At +5 dB, white noise shifts ZCR and spectral shape descriptors far more than the other conditions (Figure 4). This offers a concrete DSP explanation for the SVM's extreme white-noise sensitivity without treating feature movement as sufficient to explain every wrong label.

CNN and CLAP movement uses cosine distance between mixture and gain-matched clean embeddings. Distances are not compared absolutely across architectures. For high-overlap CNN pairs, semantic interference can produce worse accuracy despite smaller movement than matched DEMAND backgrounds. This is compatible with direction-sensitive decision boundaries, but does not prove that semantic meaning itself caused the effect. CLAP behaves differently: its high-overlap semantic mixtures generally move farther than the matched DEMAND embeddings.

CLAP also exposes the destination of movement. Under own-activity power, the same-category distractor's median rank improves from 17 in the clean control to 4, 3 and 2 at +10, +5 and 0 dB. Figure 5 shows the complete 50-class ranking diagnostic. Scores for the three highest other competing classes are retained per trial. Displacement/error AUC (Appendix) summarises association within each condition and fold, not causal explanation.

7 Deployment cost and limits

On the same local CPU with one numerical thread and batch size one, median preprocessing plus inference is approximately 11.10 ms for SVM, 2.61 ms for CNN and 98.13 ms for CLAP per five-second clip. The benchmark uses 100 repetitions after ten warm-ups on one representative clip; it is not a dataset-wide tail latency claim. Disk I/O, loading and external pretraining are excluded. CLAP prompt embeddings are cached, and its audio embedding plus class scoring is included. CPU performance is separate from MPS used for supervised training.

The compact CNN has 36,258 parameters; the loaded CLAP model has 158,334,169. Saved checkpoint size, parameter bytes and inference latency are different quantities. CLAP's 1.86-GB checkpoint must not be compared to a small supervised model file as though both contain identical kinds of state. The project measures local deployment cost, not total lifetime training energy.

Power matching leaves spectral, temporal and perceptual differences uncontrolled. Only three DEMAND environments and one semantic pair of each category type per target are sampled. Coarse ESC categories are not a direct measure of perceptual similarity. At 0 dB either event may be a reasonable description. Frozen CLAP has external pretraining advantages and potential dataset overlap. Source reuse, post-baseline analysis choices and seed variability limit generalisation. These limitations are reasons to report the protocol and failure cases precisely, not reasons to reinterpret a smaller error bar as proof of causal semantics.

8 Extension experiment status

The core and diagnostic results above are verified. The extended augmentation, multi-seed interference and prompt-ensemble results are still being generated and are not used to support the conclusions of this draft. Completed additional clean training runs at the build time are seed 5306: clean accuracy 66.40%, macro-F1 65.19%; seed 5307: clean accuracy 66.40%, macro-F1 65.53%. Their interference comparisons are reported only after evaluation and audit. The extension

compares clean, WGN-augmented and semantic-augmented supervised training. CNN seeds are 5305, 5306 and 5307. Each augmented CNN sees clean data with probability 0.5 and otherwise one of three fixed training-only views at +5, +10 or +15 dB. Banks are shared across optimisation seeds. Clean validation, architecture, training cap and four-fold refit logic are unchanged. Semantic partners come only from the current training partition; test DEMAND recordings are never used for augmentation. The SVM uses equal counts of clean examples and one balanced fixed augmented view per training clip. These interventions test transfer from ordinary acoustic corruption to competing events.

A separate fixed prompt-ensemble ablation averages text embeddings over the complete released PAT template list and reuses the exact cached CLAP audio embeddings. It does not select prompts using labels and is not called full PAT. The inspected reference implementation uses a different CLAP backend, local features and a label-based fusion-coefficient search; reproducing that search on outer test folds would invalidate this experiment's isolation. None of those steps is silently incorporated into the core CLAP reference.

9 Conclusion

Interference power alone does not describe environmental-sound classifier failure. The SVM's strongest weakness is white-noise-induced spectral-feature change, while CLAP shows substantial competition from recognisable events even when its generic-background performance remains high. Temporal overlap and power measurement materially affect interpretation; paired controls and both power protocols make that dependence visible. The smaller CNN effect should remain qualified pending replication. The practical research contribution is the combination of controlled DSP mixing, explicit reproducibility and analysis of where the classification decision moves.

Reproducibility and attribution

The repository contains custom DSP, supervised baselines, deterministic mixers, training/evaluation splits, score and embedding exports, analysis scripts and figure data. LAION-CLAP, its checkpoint and the PAT prompt list are external resources with explicit provenance. Model/data hashes and source snapshots identify the executed versions. The original proposal is retained, and its power definition change is documented above. AI-assisted implementation, checking and draft preparation were used; the report and code remain subject to author review and the course's disclosure requirements. This draft does not certify that an unverified extension or an unpublished repository update is complete.

References

[1] K. J. Piczak, "ESC: Dataset for Environmental Sound Classification," ACM Multimedia, 2015. doi:10.1145/2733373.2806390.

[2] K. J. Piczak, ESC-50 official dataset repository and documented caveats. https://github.com/karolpiczak/ESC-50.

[3] K. J. Piczak, "Environmental Sound Classification with Convolutional Neural Networks," MLSP, 2015. doi:10.1109/MLSP.2015.7324337.

[4] J. Salamon and J. P. Bello, "Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification," IEEE Signal Processing Letters, 2017. arXiv:1608.04363.

[5] J. Thiemann et al., "The Diverse Environments Multi-Channel Acoustic Noise Database (DEMAND)," POMA, 2013. doi:10.1121/1.4799597.

[6] Y. Wu et al., "Large-Scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation," ICASSP, 2023. arXiv:2211.06687.

[7] A. Seth et al., "PAT: Parameter-Free Audio-Text Aligner to Boost Zero-Shot Audio Classification," NAACL, 2025, pp. 12376–12394. doi:10.18653/v1/2025.naacl-long.616.

[8] W. Kohno et al., "Mix-CLAP: Adaptive Fusion of Knowledge-Distilled Audio Embeddings for Noise-Aware Audio-Language Models," ICASSP, 2026. Official conference record.

[9] Y. Zhang et al., "RSA-Bench: Benchmarking Audio Large Models in Real-World Acoustic Scenarios," Findings of ACL, 2026, pp. 18350–18374. doi:10.18653/v1/2026.findings-acl.913.

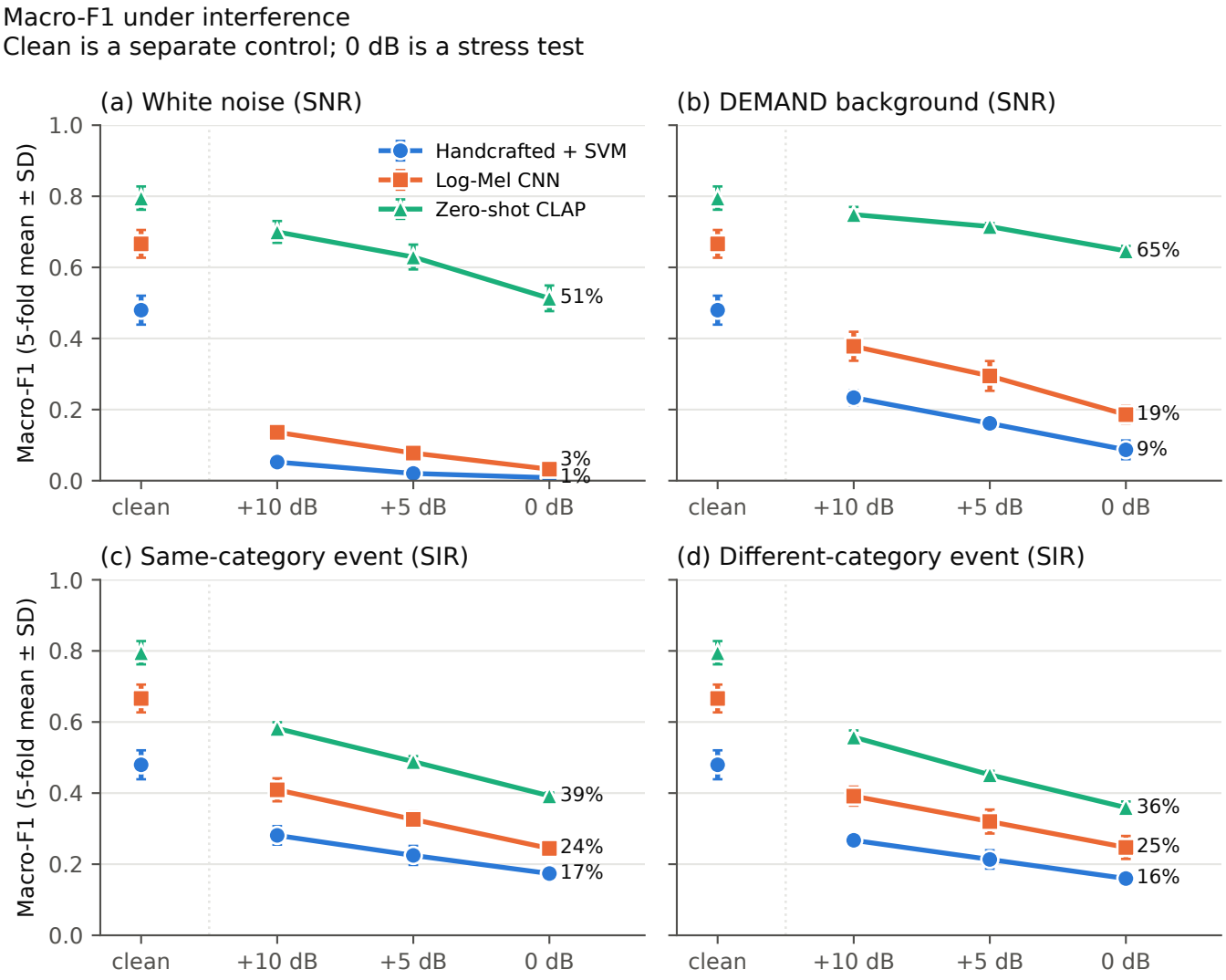


Figure 1: Core robustness curves under own-activity power. Values are five-fold macro-F1 means; error bars are sample SD, not confidence intervals. The same input mixtures are used by all systems. Zero dB is the equal-power stress test.

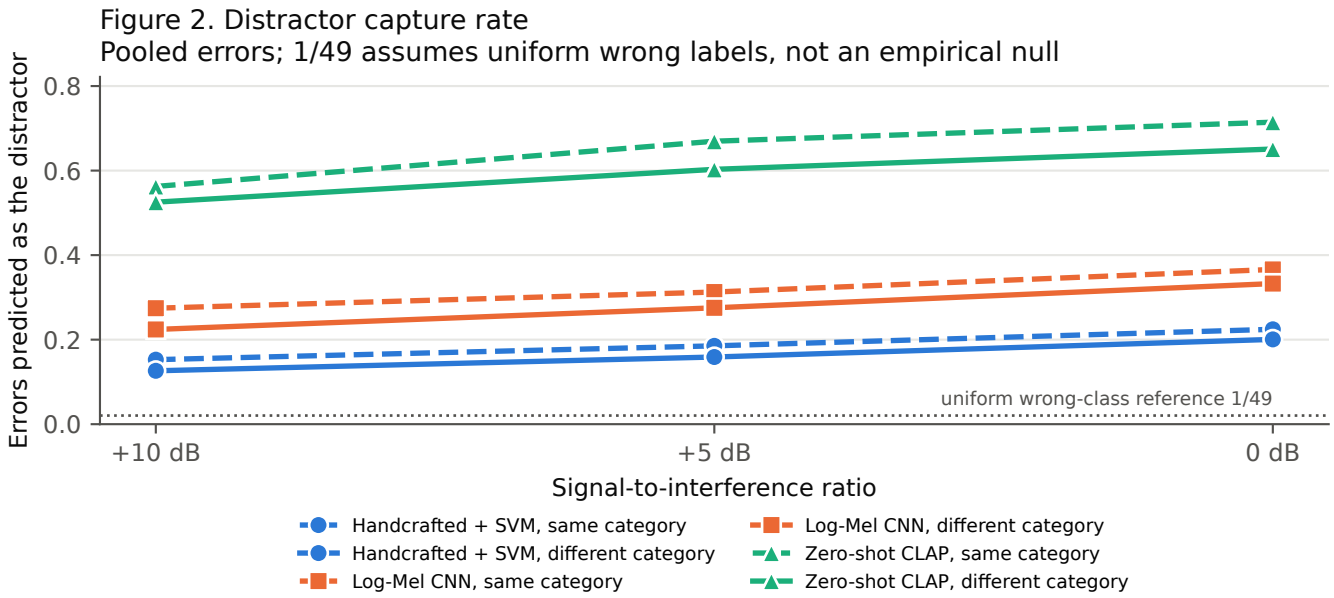


Figure 2: Distractor capture among incorrect trials. The horizontal reference is uniform wrong-label choice, not a class-conditioned null distribution. Same-category and different-category pairings are distinguished by line style.

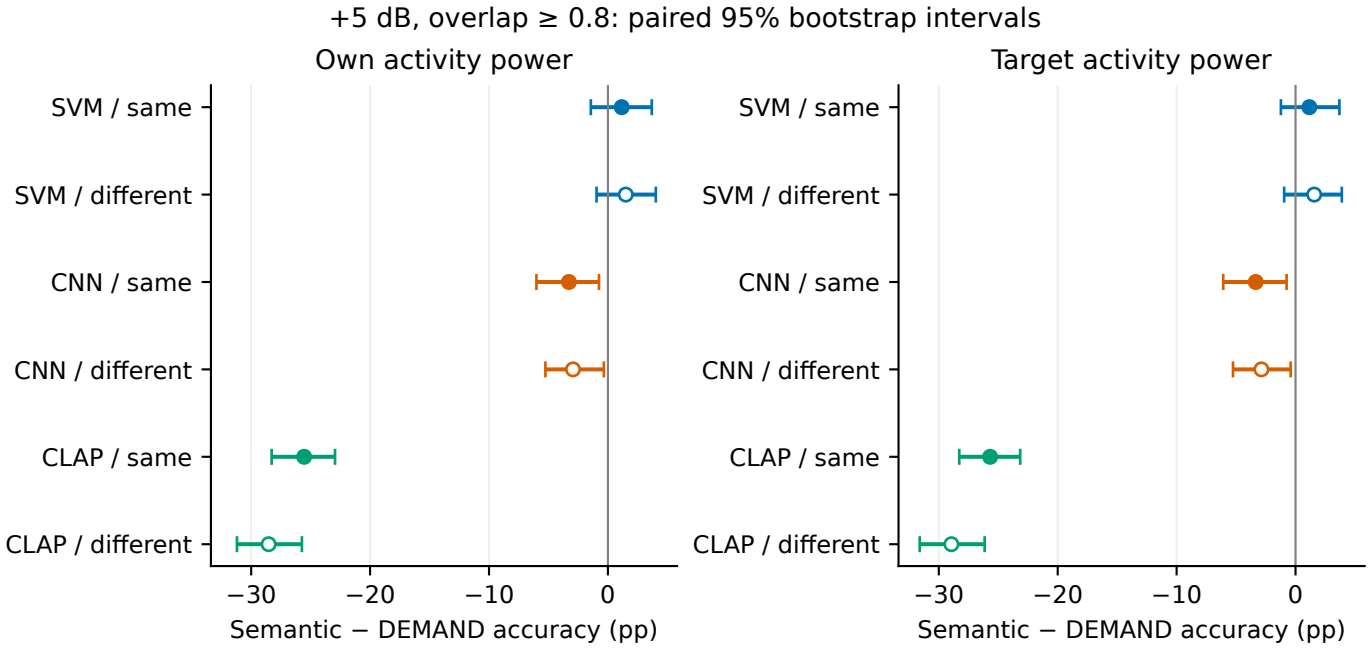


Figure 3: Semantic-minus-DEMAND accuracy on matched high-overlap targets at +5 dB. Intervals are 2,000-replicate paired target-level bootstrap intervals within folds, conditional on the trained models. They are pointwise intervals; the text separately reports multiplicity-adjusted tests.

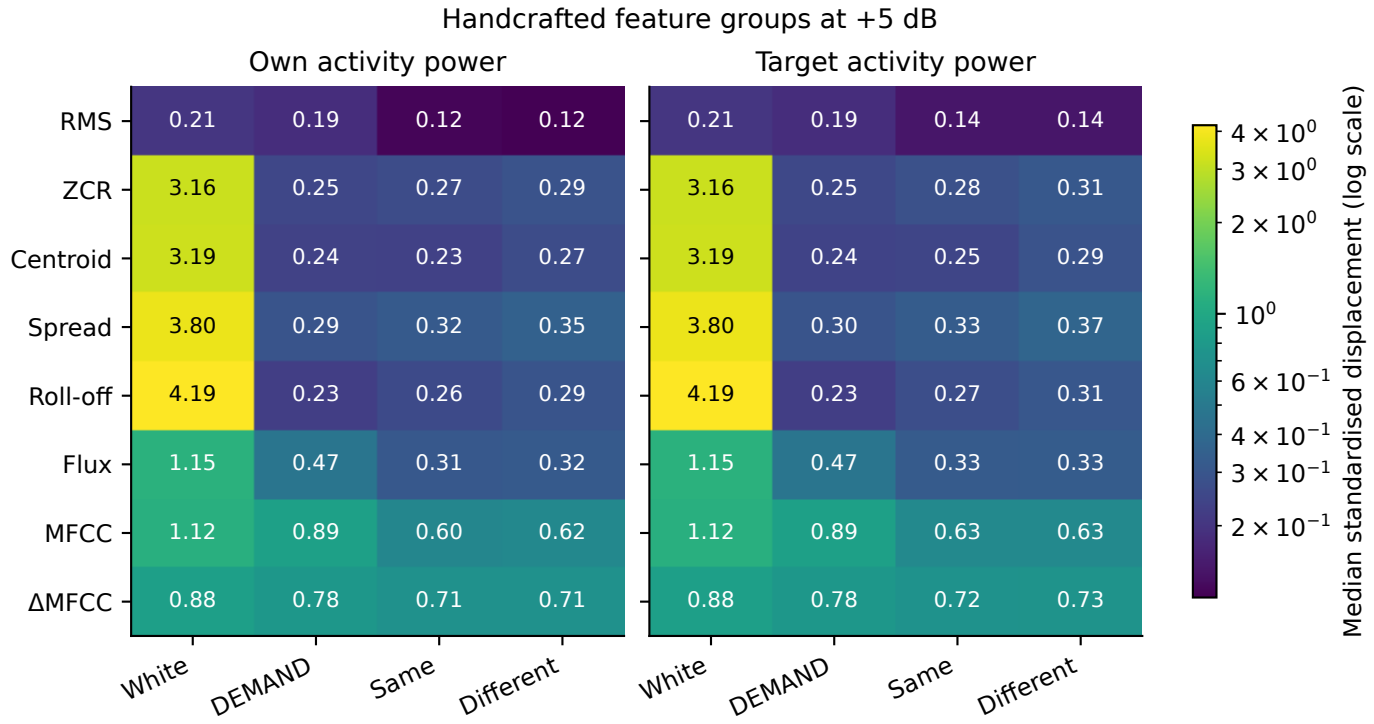


Figure 4: Median clean-control-to-mixture displacement for eight handcrafted feature groups at +5 dB. Each group uses training-fold standardisation and division by the square root of its dimension. The shared logarithmic colour scale is a display scale; annotated values are untransformed distances.

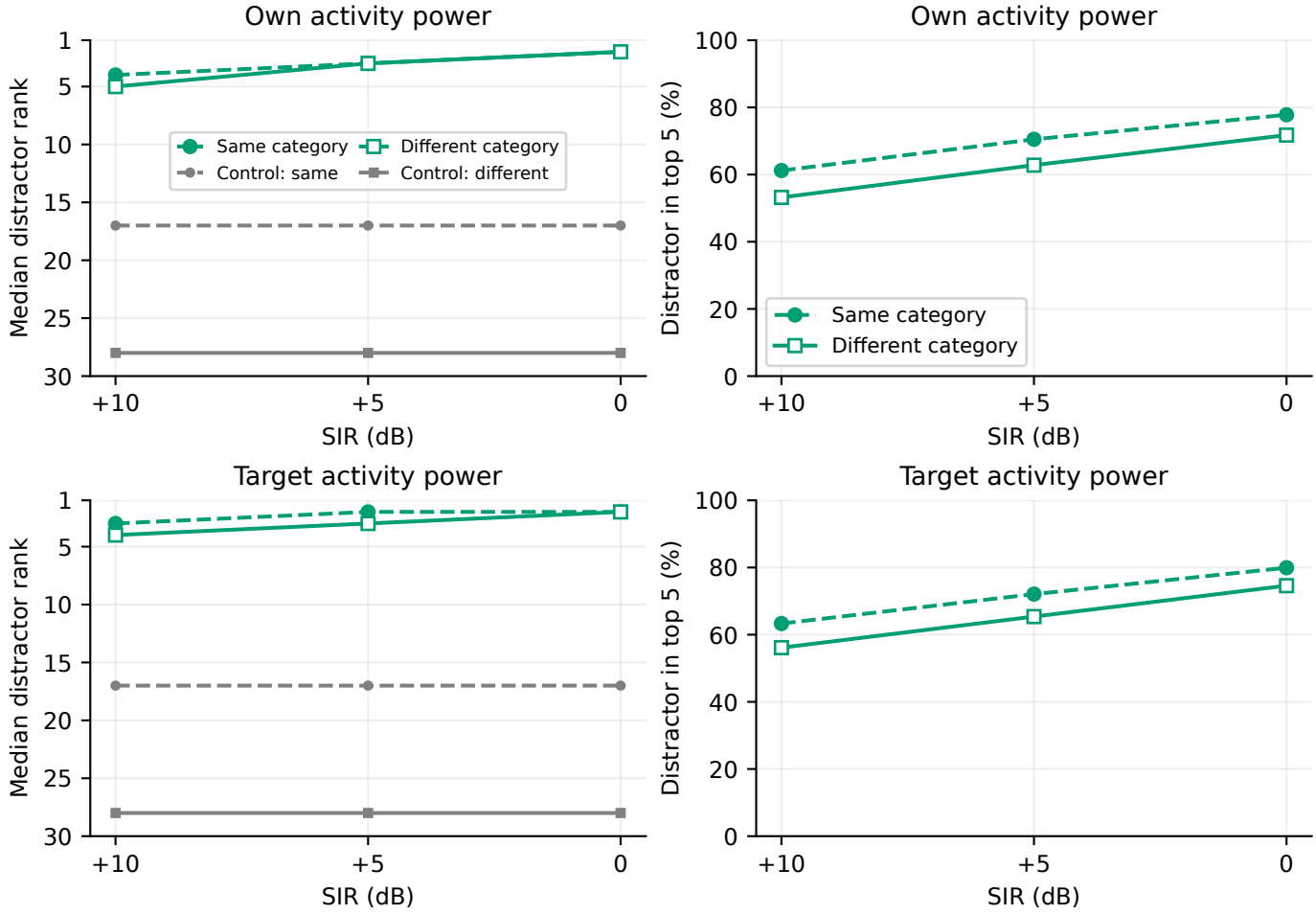


Figure 5: CLAP distractor ranking among all 50 classes. Lower rank is stronger; rank 1 is highest. Top-five fractions use all semantic trials, unlike error-conditioned capture rate. Grey curves are gain-matched clean controls.

A Supplementary figures

The following figures retain the original within-model scales and denominators. The repository also contains a complete corresponding figure set for the target power convention, each with numerical source tables and export audits.

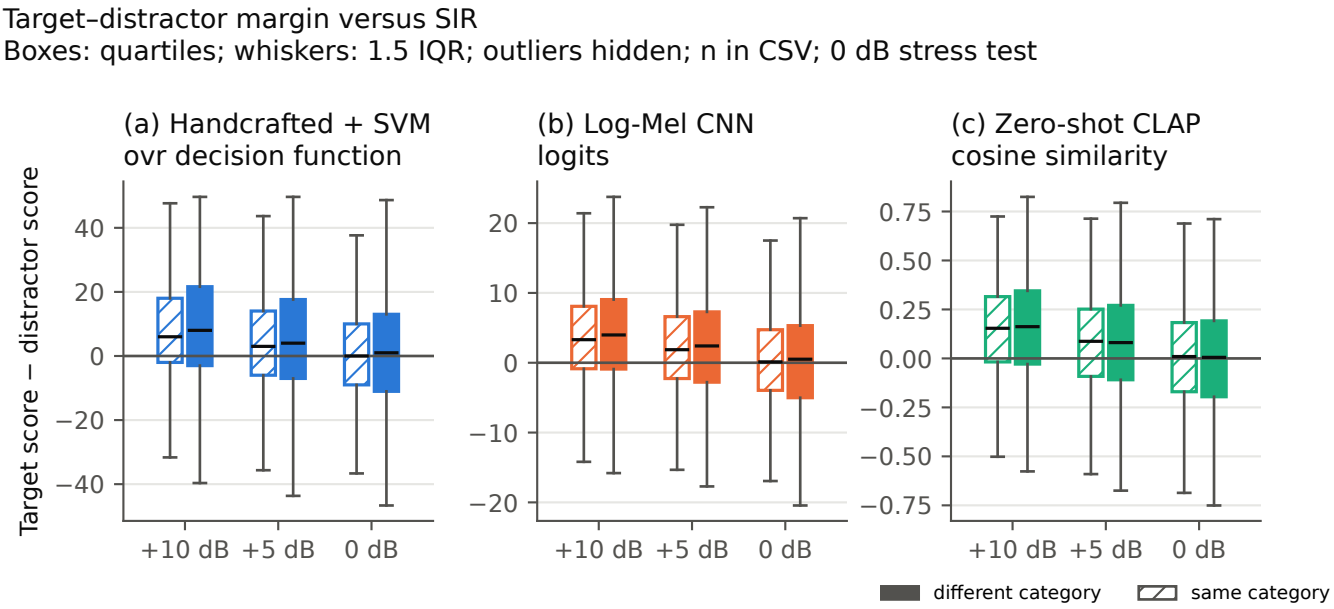


Figure 6: Target–distractor margin distributions. Scores have architecture-specific scales; comparisons are within each panel. A negative margin indicates a distractor score above the target score, not necessarily the overall predicted class.

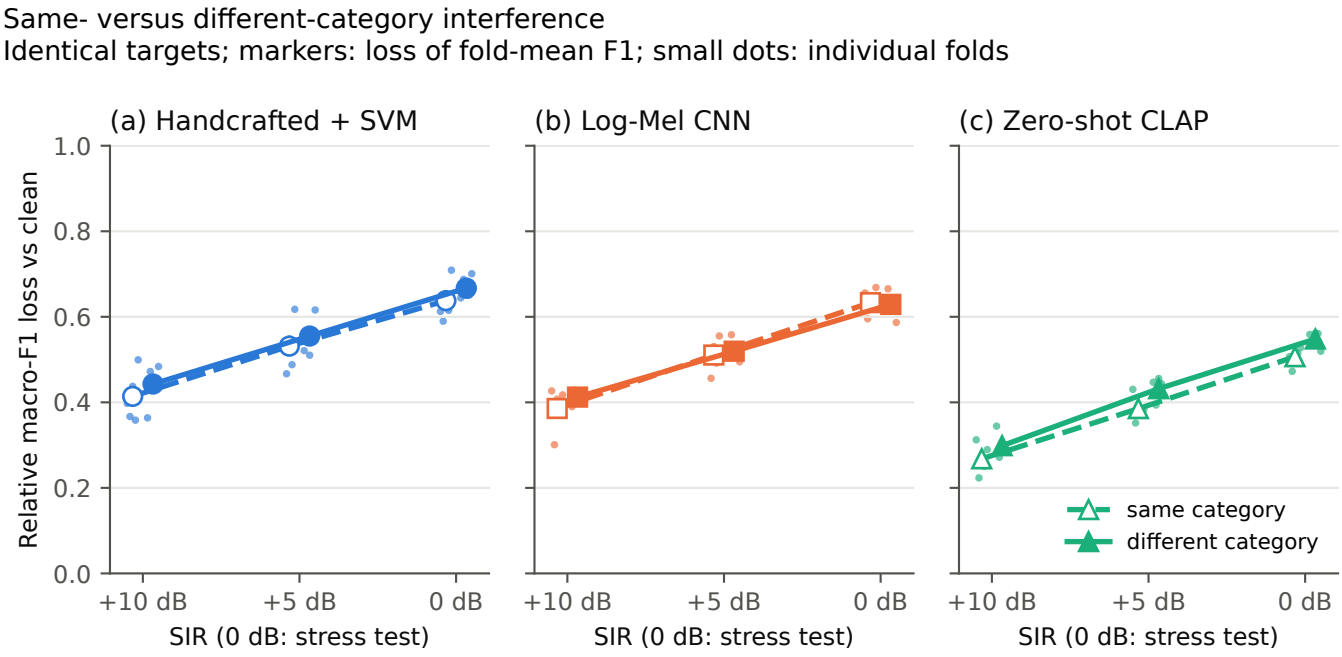


Figure 7: Same- versus different-category relative macro-F1 loss. Category relationships are coarse dataset groupings and should not be equated with a validated perceptual-distance scale.

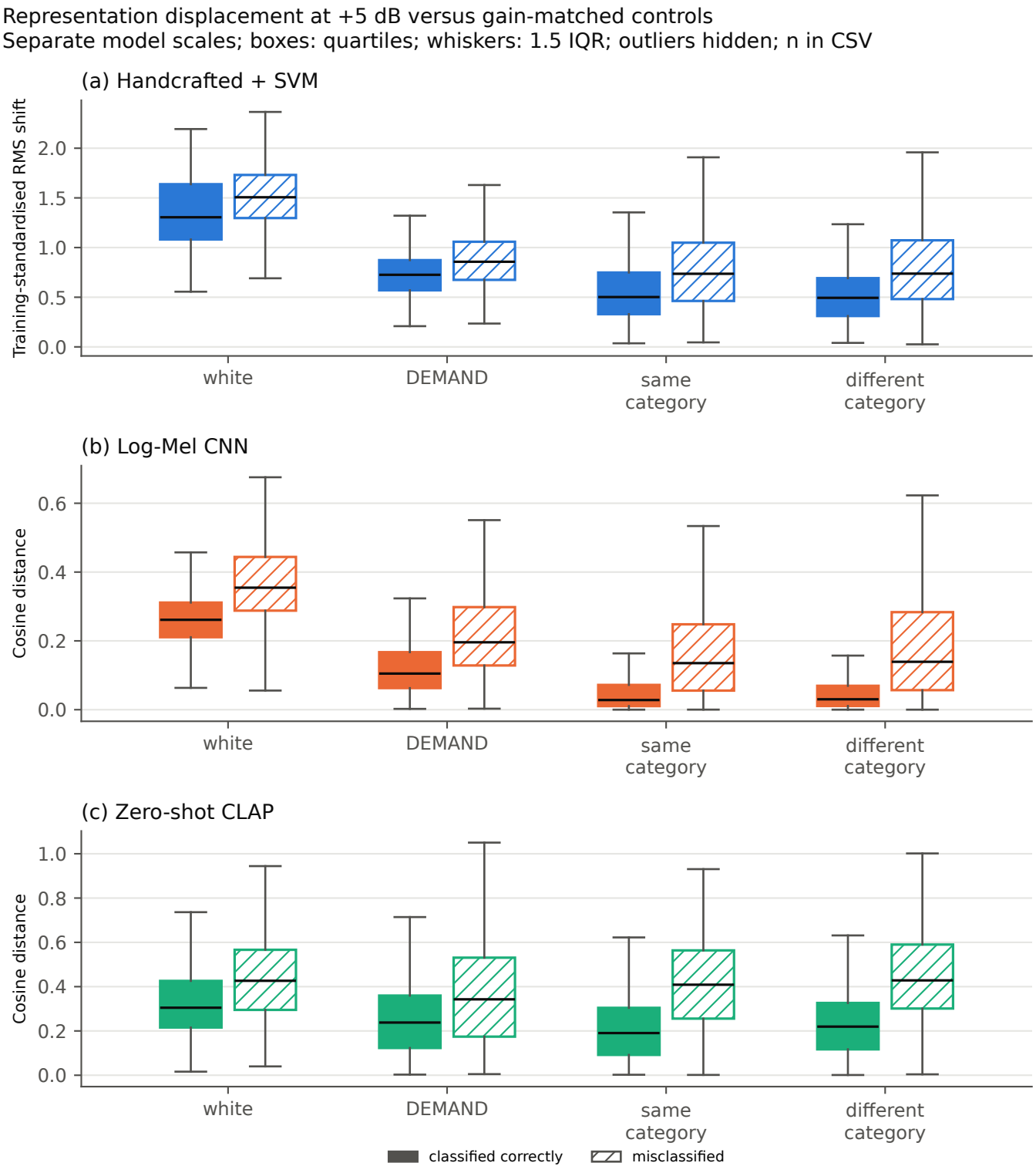


Figure 8: Representation displacement from gain-matched controls, separated by classification outcome. Handcrafted features use training-fold standardisation; CNN and CLAP use cosine distance. Absolute distances are not comparable between architectures.

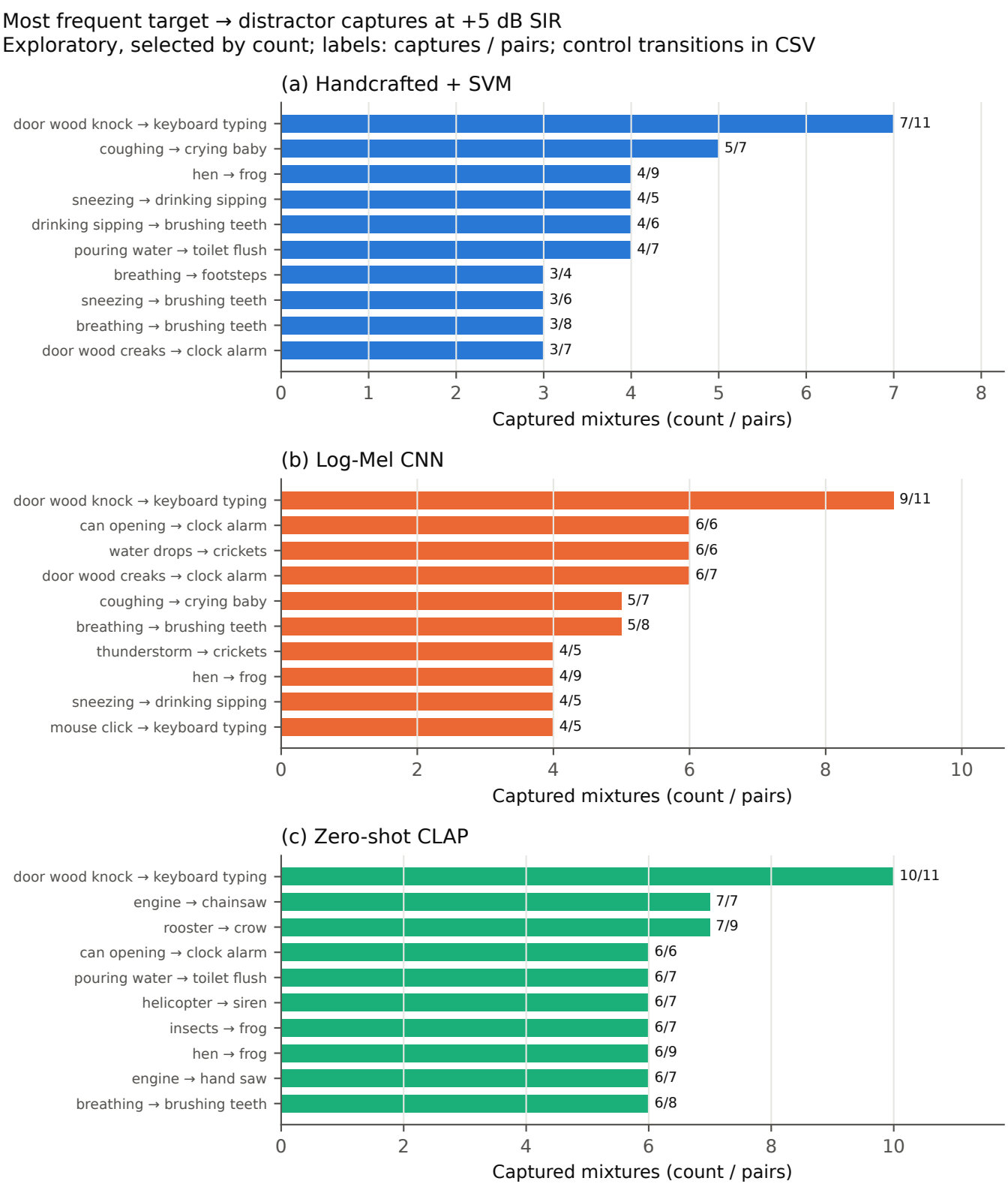


Figure 9: Observed target-to-distractor capture flows. Labels report captured trials relative to the available pair count. Small denominators make these illustrative examples rather than reliable estimates of class-pair risk.

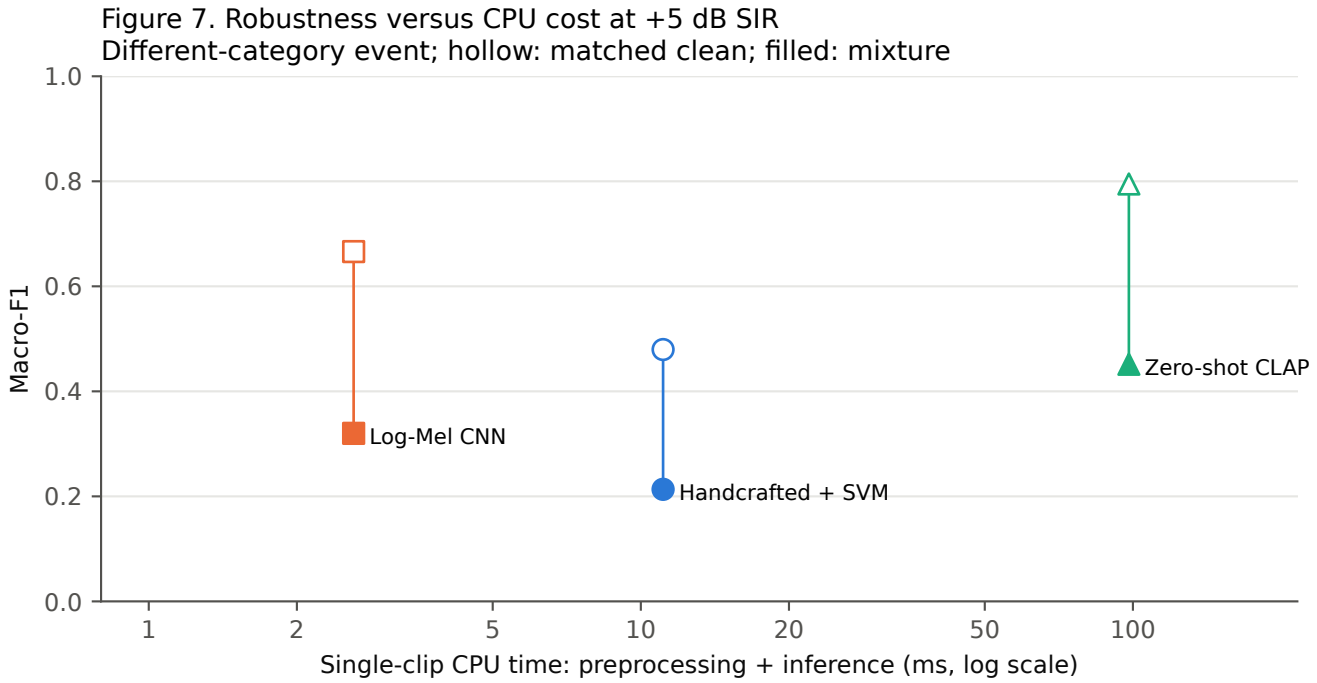


Figure 10: Own-protocol robustness and CPU cost at +5 dB different-category interference. Timing is a single-clip, batch-one microbenchmark with cached CLAP text embeddings; open markers show clean macro-F1. External pretraining cost is excluded.

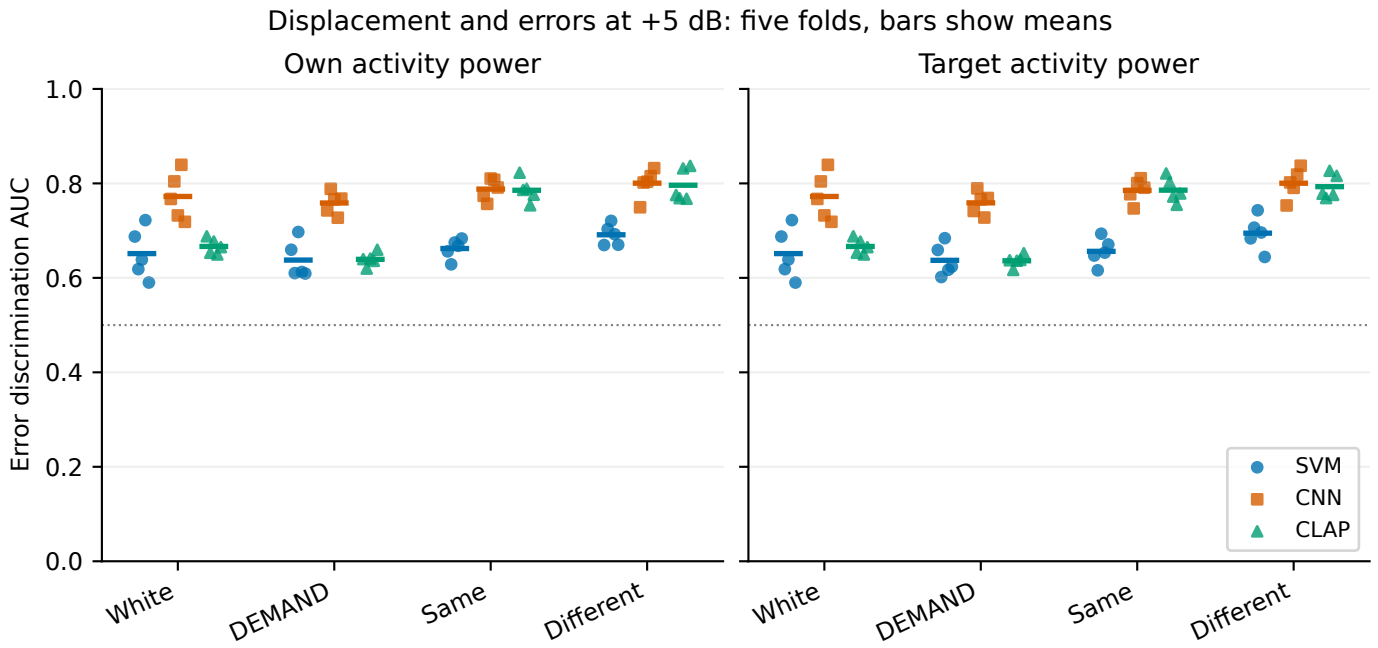


Figure 11: Association between representation displacement and classification error at +5 dB. Dots show the five official folds and short bars their means. AUC above 0.5 indicates larger movement tends to accompany errors; this is descriptive, not a causal semantic-effect estimate.

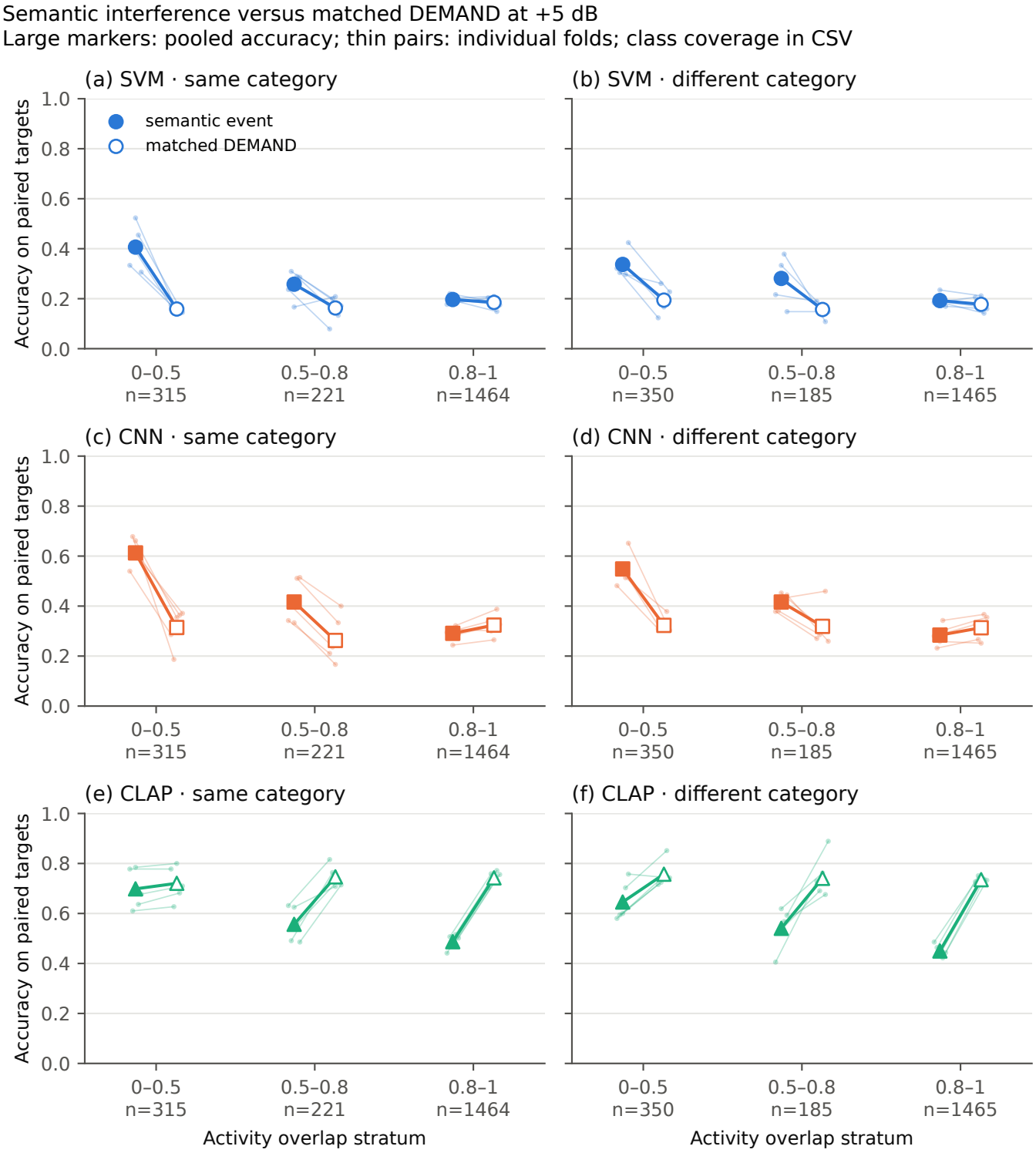


Figure 12: Overlap-stratified semantic versus DEMAND comparisons on matched target sets. Read accuracy together with class coverage; macro-F1 with all 50 labels is sensitive to absent classes in small strata.